Forget the distant past: training windows and window combination for automatic ARIMA and exponential smoothing on long time series

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Abstract

Statistical forecasting methods are usually fitted to the whole of the available history. For long series, with thousands of observations, this assumes one model with fixed parameters holds throughout, which is rarely plausible. We study whether automatic ARIMA and exponential smoothing (ETS), as implemented in the `aeon` toolkit, forecast better when fitted to a window of recent data, and how to choose or combine windows automatically. On LSSH-99, a collection of 99 long series of up to 10,000 observations, with one-step rolling-origin evaluation, fixed windows of 300–1,000 observations reduce the MAE of AutoARIMA and AutoETS by 6–8% relative to the whole series. Combining the forecasts from a geometric grid of windows is more reliable than choosing one: the median over windows reduces error by 11–13%, and weighting windows by their recent out-of-sample error by 12–15%, improving about 70% of series. On the 5,070 M4 series with at least 500 observations, under the standard M4 protocol, the gains are smaller (OWA 0.920 to 0.894–0.905 for `AutoARIMA`, 0.945 to 0.922–0.926 for `AutoETS`), concentrated on hourly, monthly and weekly series, with no effect on daily series. The median over windows, and the median of the half of the windows with the lowest out-of-sample error, are close to the best scheme on both benchmarks; weights from in-sample fit, Akaike-style weights and selection of the best-fitting window are no better, or worse, on M4, consistent with the forecast combination puzzle. We propose a simple wrapper that turns any forecaster into a window-combining forecaster.

1 Introduction

Automatic statistical forecasters such as `auto.arima` and `ets` (Hyndman and Khandakar, 2008; Hyndman et al., 2002, 2008) remain strong benchmarks: in the M4 competition combinations of them were among the most accurate methods (Makridakis et al., 2020). They are almost always fitted to the whole of the available history. For the short series that dominate the M competitions this is sensible, since there is little data to spare. For long series it is less obvious. A series of 10,000 hourly observations covers more than a year; its level, seasonal profile and dynamics are unlikely to be constant over that period, and a model estimated on all of it averages over regimes that may no longer apply. Practical advice recognises this: Hyndman and Athanasopoulos (2021) note that "most time series models do not work well for very long time series" and suggest, for short-horizon forecasts, to "throw away the earliest observations and only fit a model to the most recent observations". They give no rule for how many to keep.

The question of how much history to use has a long history in econometrics, where it is framed as the choice of an estimation window under structural breaks (Pesaran and Timmermann, 2007) and has led to methods that combine forecasts across estimation windows (Pesaran and Pick, 2011). That work concerns mostly regression and autoregressive models with a fixed specification applied to macroeconomic and financial series. Here we ask the same question of automatic ETS and ARIMA, which select a model on every fit, on benchmark collections of long univariate

series, and with the interaction between windows and the seasonal adjustment used in forecasting practice.

We make three contributions.

1. An evaluation of training windows for automatic ARIMA and ETS on LSSH-99, a collection of 99 long series (Banwell and Bagnall, 2026), and on the long series of M4, showing that fitting all the data is generally no better, and often clearly worse, than fitting recent data.

2. A comparison of ways to choose a window or combine forecasts across windows: per-series and rolling selection, means, medians, trimmed means, weights from in-sample fit normalised for window length, Akaike-style weights, and weights from recent out-of-sample error.

3. A simple forecaster wrapper derived from these results (its implementation and evaluation as an ordinary forecaster are in progress), and a store of all window forecasts that allows further combination schemes to be evaluated without refitting.

2 Related work

Choosing the estimation window. Pesaran and Timmermann (2007) frame the choice as a trade-off between the bias from using data from before a break and the variance from a shorter sample, and propose cross-validation methods to choose a single window together with methods that weight or pool forecasts from windows of different lengths. Inoue et al. (2017) choose the rolling window to minimise an estimate of the forecaster's quadratic loss, by analogy with bandwidth selection in nonparametric estimation. Clark and McCracken (2009) show that combining forecasts from an expanding (recursive) and a rolling window is at least as accurate as either. Rossi and Inoue (2012) make the related point for evaluation: tests of predictive ability depend on the window, and repeating them over windows invites data snooping.

Combining across windows. Pesaran and Pick (2011) average one model's forecasts over estimation windows and find lower bias and RMSFE than any single window for all but the smallest breaks. Tian and Anderson (2014) weight windows by the estimated probability that each point is the most recent break. Assenmacher-Wesche and Pesaran (2008) find averaging over windows at least as effective as averaging over models for Swiss macroeconomic forecasts, with little gain from weighting schemes over equal weights, and Jungmittag (2016) find window-averaged SARIMA forecasts of monthly air travel generally better than single windows. Most recently, Petersen et al. (2026) compare training window selection methods for regression and random forests on non-stationary psychological time series: a simple average across windows and a super learner ensemble perform best, while selecting a single window by cross-validation performs poorly, and the expanding window is best only when the coefficients are stable.

Down-weighting instead of truncating. Pesaran et al. (2013) derive the weights on past observations that minimise mean squared forecast error under breaks, close to exponential under continuous change; Giraitis et al. (2013) choose the discount rate of rolling and exponentially weighted estimators by cross-validation. Outside econometrics, adaptive windowing for drifting data streams (Bifet and Gavaldà, 2007) and dynamic ensembles that weight forecasters over time (Cerqueira et al., 2017) address related problems.

Forecast combination. Equal weights are hard to beat: estimated weights rarely improve on the simple average (Stock and Watson, 2004), because of the error in estimating them (Smith and Wallis, 2009; Claeskens et al., 2016). Trimmed and Winsorised means are slightly more accurate and more robust than the mean (Jose and Winkler, 2008). Kolassa (2011) combines ETS models

fitted to the same series with Akaike weights, which beat selection by an information criterion. Bagging ETS over bootstrapped series (Bergmeir et al., 2016) improves on ETS, mainly through the uncertainty in model selection (Petropoulos et al., 2018), a mechanism that also operates when an automatic forecaster selects different models on different windows.

What is new here. We have not found an evaluation of window selection or combination for automatic ETS and ARIMA on benchmark collections of long series, of the median across windows, or of the interaction between windows and seasonal adjustment; nor a forecasting library that offers combination across estimation windows as a component.

3 Methods

3.1 Forecasters

We use four forecasters from `aeon` (Middlehurst et al., 2024), at the code released in version 1.7: `AutoARIMA` (stepwise order search over non-seasonal ARIMA models, fitted by conditional sum of squares), `AutoETS` (selection over error, trend and seasonal components by AIC), and two fixed models, `ARIMA`(1,0,1) and simple exponential smoothing `ETS`(A,N,N), the toolkit defaults. As in the M4 benchmarks, the non-seasonal `AutoARIMA` and `ARIMA` are fitted to series deseasonalised by classical multiplicative decomposition when a 90% autocorrelation test finds seasonality, and the forecasts are reseasonalised; `AutoETS` is given the seasonal period; `ETS`(A,N,N) is fitted to the raw series.

3.2 Training windows

Given a series $y_1, \ldots, y_t$ observed up to the forecast origin t, the forecaster fitted to window w uses $y_{t-w+1}, \ldots, y_t$; $w = 0$ denotes the whole series. We use the grid

$$W = \{50, 100, 200, 300, 500, 750, 1000, 1500, 2000, 3000, 5000, 7500\}$$

plus the whole series, skipping windows at least as long as the history. Let $\hat{y}_{t+h}^{(w)}$ be the forecast from window w.

3.3 Choosing a window

Validation selection chooses, per series, the window with the lowest error on a block of origins before the test period (LSSH-99), or on the last h training observations (M4). *Rolling* selection chooses, at each origin, the window with the lowest mean one-step error over the previous 30 origins. *Global* selection chooses one window for all series.

3.4 Combining windows

A combination forecasts $\sum_w \lambda_w \hat{y}_{t+h}^{(w)}$ with weights $\lambda_w \geq 0$ summing to one, or a robust location of the window forecasts.

- *Mean*, *median*, and the *trimmed mean* (20% from each end).
- *In-sample fit.* With $e_s^{(w)}$ the in-sample one-step residuals of the fit to window w, we use per-point scores so that windows of different lengths are comparable: the mean squared residual MSE_w, and the relative MAE r_w, the mean absolute residual divided by the mean absolute one-step change over the same points (the in-sample error of the naive forecast), which also removes the local scale of the series. Weights are $\lambda_w \propto \mathrm{MSE}_w^{-1}$, $\propto r_w^{-1}$ or $\propto r_w^{-2}$.

- *Akaike-style.* The AIC is not comparable across samples of different size, so we scale the per-point Gaussian log-likelihood to a common length $n_0 = 100$: $\lambda_w \propto \exp\{-\frac{n_0}{2}(\log \mathrm{MSE}_w - \min_v \log \mathrm{MSE}_v)\}$.
- *Best in-sample window*: the window with the smallest r_w.
- *Out-of-sample error.* $\lambda_w \propto \mathrm{OOS}_w^{-1}$, with OOS_w the mean squared one-step error of window w at the previous 30 origins (LSSH-99) or on the M4 holdout; and the median of the better half of windows by OOS_w.
- *Minimum window.* The mean, median, inverse in-sample MSE and inverse out-of-sample MSE restricted to windows of at least 300 observations (and the whole series).

For multiplicative-error ETS models the residuals returned are relative errors; we convert them to the scale of the data by multiplying by the fitted values.

4 Experimental design

LSSH-99. The collection of Banwell and Bagnall (2026): 99 long series (1,008 to 10,000 observations), most from the Monash archive (Godahewa et al., 2021), including 57 long M4 series, with seasonal periods from the Monash conventions (48 half-hourly, 24 hourly, 7 daily, 144 ten-minute, 1440 minutely, 52 weekly, 12 monthly). Each forecaster is refitted at each of the last 60 observations and forecasts one step ahead; origins 1–30 form a validation period and origins 31–60 the test period, the same test origins as in the evaluation of `aeon`'s forecasters by Banwell and Bagnall (2026). Per series, the MAE over the test origins is divided by that of the naive (last value) forecast; methods are compared by the geometric mean of this ratio over 98 series (one series is constant, and on one the naive forecast is exact), on the 36 sub-daily series and the 62 daily-or-slower series. For each method against the whole series we count series better and worse and apply a Wilcoxon signed-rank test to the log ratios, Holm-corrected within each forecaster. The whole-series forecasts are identical to those of the reference runs.

M4 long series. The 5,070 M4 series (Makridakis et al., 2020) with at least 500 training observations (2,946 daily, 1,403 monthly, 414 hourly, 252 weekly, 43 quarterly, 12 yearly; median length 1,430), with the standard protocol: one origin at the end of the training series, the competition horizons, sMAPE and MASE averaged over the forecast points, and OWA relative to the published Naive2 forecasts on the same series. Each window is also fitted to the training series without its last h observations, forecasting them, to provide a holdout for selection and for out-of-sample weights.

Reproducibility. All forecasts are stored per window, with the in-sample fit scores, so that any further combination can be evaluated without refitting. Code, results and the store are in the accompanying repository [TODO: URL].

5 Results

5.1 Does a window beat the whole series?

Table 1 gives the LSSH-99 relative MAE of each fixed window. For `AutoARIMA`, `AutoETS` and `ARIMA`(1,0,1), windows of 300–1,000 observations beat the whole series by 6–8%; very short windows (50–200) are no better than the whole series, and the error rises back towards the whole-series value as the window grows beyond 2,000. Simple exponential smoothing is almost unaffected (1–2%). No single fixed window is significantly better than the whole series after correction for the 13 windows tested.

Table 1: LSSH-99: relative MAE (geometric mean over 98 series of MAE divided by the naive MAE; below 1 is better than naive) of each forecaster fitted to the last w observations at each of 30 one-step origins. Bold: best per row.

	Training window (observations)												
Forecaster	Whole	50	100	200	300	500	750	1000	1500	2000	3000	5000	7500
`AutoARIMA`	0.918	0.907	0.924	0.884	0.861	**0.846**	0.847	0.865	0.873	0.865	0.882	0.889	0.909
`AutoETS`	0.927	1.024	0.987	0.917	0.909	0.880	0.868	**0.859**	0.879	0.876	0.905	0.910	0.915
`ARIMA`(1,0,1)	0.932	0.929	0.946	0.936	**0.862**	0.866	0.864	0.866	0.875	0.882	0.914	0.918	0.931
`ETS`(A,N,N)	1.023	1.016	1.007	1.005	**1.005**	1.007	1.006	1.008	1.026	1.026	1.025	1.024	1.021

Table 2: LSSH-99: relative MAE of the whole series, the best fixed window (in brackets), window selection and window combination. Oracle: best window per series chosen on the test errors (not a method).

	Whole	Best fixed	Selection		Combination			
Forecaster	series	window	validation	rolling	mean	median	inv. OOS MSE	Oracle
`AutoARIMA`	0.918	0.846 (500)	0.813	**0.800**	0.834	0.813	0.803	0.745
`AutoETS`	0.927	0.859 (1000)	0.861	0.808	0.818	0.804	**0.789**	0.734
`ARIMA`(1,0,1)	0.932	0.862 (300)	0.824	**0.823**	0.881	0.846	0.840	0.787
`ETS`(A,N,N)	1.023	**1.005** (300)	1.013	1.006	1.014	1.010	1.012	0.990

5.2 Choosing or combining windows

Table 2 compares choosing a window with combining windows. Choosing per series on a validation block beats the best fixed window for `AutoARIMA` and `ARIMA`(1,0,1) but not for `AutoETS`; rolling selection, which re-chooses the window at each origin from the previous 30 errors, gains 12–13% for `AutoARIMA` and `AutoETS`. Combination is at least as good and more reliable: the median over windows improves on the whole series by 11.5% (`AutoARIMA`) and 13.2% (`AutoETS`), for 66 and 73 of the 98 series, both significant, and it is better than the mean for every forecaster. Weighting the windows by their recent out-of-sample error is better still for `AutoETS` (0.789 against 0.927 for the whole series, 72 series better). The oracle, which picks the best window per series with knowledge of the test errors, shows that more is available (0.73–0.75).

5.3 Weighting schemes

Table 3 gives every combination scheme. Three findings stand out. First, the gains over the median are small: the best weighted schemes improve on it by 1–2%, as the forecast combination puzzle (Smith and Wallis, 2009; Claeskens et al., 2016) leads one to expect. Second, the schemes that rely most on in-sample fit quality, selecting the window with the best in-sample fit and Akaike-style weights (which concentrate almost all weight on the best-fitting window), are the weakest weighted schemes for `AutoARIMA` and `AutoETS`, and not significantly better than the whole series for `AutoARIMA`: in sample, a model fitted to fewer points fits them more closely. Third, inverse in-sample MSE weights do well (0.799 for `AutoARIMA`, 0.790 for `AutoETS`). They are far from selection: the largest weight is a median of about 0.2 (equal weights give 0.08), and they tilt the combination mildly towards shorter windows, over which the series has been calmer (weighted geometric mean window 480 against 597 under equal weights for `AutoARIMA`, 538 for `AutoETS`). Unlike the relative MAE, this score is not scale-free, so it rewards windows covering low-volatility periods, which on these data acts as a mild recency weighting.

Table 3: LSSH-99: relative MAE of each combination scheme, with the number of series better / worse than the whole series. *: significantly different from the whole series (Wilcoxon signed-rank, Holm-corrected, $p < 0.05$). Bold: best per forecaster.

Combination	`AutoARIMA`	`AutoETS`	`ARIMA`(1,0,1)	`ETS`(A,N,N)
Whole series	0.918	0.927	0.932	1.023
Mean	0.834* (66/32)	0.818* (68/30)	0.881 (57/41)	1.014 (38/47)
Median	0.813* (66/32)	0.804* (73/25)	0.846* (60/38)	1.010 (27/24)
Trimmed mean (20%)	0.824* (68/30)	0.806* (74/24)	0.870* (58/40)	1.012 (38/35)
Inverse in-sample MSE	**0.799*** (62/36)	0.790* (72/26)	0.844 (59/39)	1.010 (35/55)
Inverse in-sample relative MAE	0.815* (69/29)	0.803* (69/29)	0.850* (59/39)	1.013 (39/46)
Inverse squared in-sample relative MAE	0.808* (70/28)	0.799* (71/27)	0.845* (59/39)	1.012 (38/46)
Akaike-style weights ($n_0 = 100$)	0.825 (49/49)	0.831* (61/37)	0.852 (58/40)	1.007 (37/54)
Best in-sample window	0.836 (47/51)	0.820* (59/39)	0.851 (55/43)	1.010 (27/33)
Inverse out-of-sample MSE	0.803* (69/29)	**0.789*** (72/26)	0.840* (59/39)	1.012 (39/53)
Median of better half	0.805* (65/33)	0.796* (73/25)	**0.837*** (59/39)	**1.005** (27/28)
Mean, $w \geq 300$	0.844* (63/35)	0.842* (67/31)	0.880* (61/37)	1.016 (43/29)
Median, $w \geq 300$	0.836* (63/35)	0.837* (74/24)	0.866* (60/38)	1.019 (30/20)
Inverse in-sample MSE, $w \geq 300$	0.835* (64/34)	0.830* (71/27)	0.863* (62/36)	1.013 (44/46)
Inverse out-of-sample MSE, $w \geq 300$	0.832* (64/34)	0.827* (69/29)	0.862* (61/37)	1.014 (48/41)

Table 4: LSSH-99 by sampling frequency (seasonal period): relative MAE of the whole series, the median over windows and inverse out-of-sample MSE weights.

		`AutoARIMA`			`AutoETS`		
Sampling (period)	Series	whole	median	inv. OOS	whole	median	inv. OOS
Daily (7)	27	1.072	1.057	1.062	1.086	1.072	1.080
Monthly (12)	18	0.992	0.940	0.879	0.941	0.904	0.895
Hourly (24)	21	0.774	0.543	0.533	0.825	0.638	0.602
Half-hourly (48)	7	0.733	0.750	0.743	0.741	0.512	0.512
Weekly (52)	17	0.822	0.771	0.779	0.817	0.706	0.674
10-minute (144)	5	0.914	0.910	0.912	1.109	0.998	0.944
Minutely (1440)	5	1.492	0.718	0.746	1.192	0.641	0.728

5.4 Where windows help

Table 4 breaks the LSSH-99 results down by sampling frequency. The gains are concentrated on sub-daily data: hourly series (`AutoARIMA` 0.774 to 0.543 with the median) and minutely wind power, where the whole-series fits are worse than naive (1.49 and 1.19) and windows bring them to 0.64–0.75. `AutoETS` also gains strongly on half-hourly series (0.741 to 0.512), `AutoARIMA` does not. On weekly and monthly series windows help moderately; on daily series, which are close to random walks, they make little difference. A likely mechanism for the hourly series is that deseasonalisation and the seasonal models estimate the daily profile from the window, so the profile is local rather than averaged over a year.

5.5 M4 long series

Tables 5 and 6 give OWA on the 5,070 long M4 series. The gains are much smaller than on LSSH-99: the best schemes reduce OWA from 0.920 to 0.892–0.898 for `AutoARIMA` and from 0.945 to 0.922–0.926 for `AutoETS`, about 2–3%. They are concentrated where LSSH-99 found them: hourly (`AutoARIMA` 0.692 to 0.578–0.648 for the schemes in Table 5 other than the mean), monthly (0.917 to 0.850–0.876) and weekly series (0.878 to 0.829–0.860). Daily series, 58% of the subset and close to random walks, show no effect for any method. The M4 series are also shorter (median 1,430 against up to 10,000), and the forecasts multi-step.

Very short windows are harmful on hourly series. Below 300 observations the M4 seasonality

Table 5: M4 series with at least 500 training observations: OWA relative to Naive2 by frequency. Holdout selection uses windows of at least 300 observations; the combinations use all windows.

	Yearly	Quarterly	Monthly	Weekly	Daily	Hourly	All
`AutoARIMA`							
Whole series	1.019	0.998	0.917	0.878	1.007	0.692	0.920
Holdout selection ($w \geq 300$)	1.010	0.990	0.876	0.860	1.007	0.578	0.897
Mean	1.005	0.971	0.858	0.829	1.001	1.221	0.975
Median	1.011	0.973	0.855	0.843	1.002	0.648	0.905
Median of better half	0.994	0.987	0.863	0.851	1.003	0.610	0.898
Inverse holdout MSE	0.993	0.976	0.856	0.837	1.003	0.670	0.910
Inverse holdout MSE, $w \geq 300$	1.002	0.993	0.874	0.844	1.002	0.600	0.894
Inverse in-sample MSE	0.993	0.944	0.850	0.841	0.996	0.697	0.912
`AutoETS`							
Whole series	0.990	0.939	0.869	0.963	1.002	0.788	0.945
Holdout selection ($w \geq 300$)	0.955	0.945	0.857	0.928	0.999	0.768	0.934
Mean	0.988	0.913	0.844	0.934	0.999	1.697	1.027
Median	0.992	0.934	0.836	0.932	1.003	0.750	0.926
Median of better half	0.958	0.938	0.842	0.924	1.002	0.733	0.924
Inverse holdout MSE	0.982	0.899	0.849	0.920	0.998	0.734	0.926
Inverse holdout MSE, $w \geq 300$	0.982	0.937	0.850	0.931	0.999	0.719	0.922
Inverse in-sample MSE	0.970	0.899	0.842	0.957	1.000	0.844	0.943

test, which considers lags up to $10 \log_{10} n$, no longer reaches lag 24, so the non-seasonal models are fitted without seasonal adjustment, and `AutoETS` fits seasonal models to as few as 50 points. These windows ruin the mean over windows (`AutoETS` OWA 1.027, worse than the whole series) but not the median (0.926): the median does not depend on choosing the minimum window well. Restricting the windows to at least 300 observations rescues the mean on M4 (`AutoARIMA` 0.897) but hurts every scheme on LSSH-99 (`AutoETS` median 0.804 to 0.837), where short windows contribute to the one-step forecasts.

The weighting schemes behave differently on the two benchmarks. Weights from in-sample fit, which were among the best on LSSH-99, are no better than the whole series on M4 for `AutoETS` (inverse in-sample MSE 0.943, relative MAE 0.955, best in-sample window 0.960, Akaike-style 0.978). Weights from the holdout error remain among the best (0.910 and 0.926 over all windows, 0.894 and 0.922 over windows of at least 300). The median of the better half of the windows by out-of-sample error is among the four best schemes in all four comparisons (LSSH-99 0.805 and 0.796; M4 0.898 and 0.924 for `AutoARIMA` and `AutoETS`); the plain median and inverse out-of-sample MSE weights are within 2% of the best scheme in each comparison.

5.6 Cost

A window of 500–1,000 observations costs 2–4 times less to fit than the whole series on LSSH-99 (median `AutoARIMA` fit 0.20–0.35 s against 0.84 s; `AutoETS` 0.33–0.53 s against 1.10 s). Fitting every window costs about 4 times the whole-series fit (median over series and origins 3.7–4.0, interquartile range about 3.2–5.0), since most of the windows are short; rolling selection and out-of-sample weights additionally need the window forecasts at previous origins, which in a rolling evaluation are available at no extra cost.

6 A window-combining forecaster

The results suggest a simple wrapper, Algorithm 1. It turns any forecaster into one that fits clones to a grid of recent windows and combines their forecasts with the median, which needs no extra fits and is within 2% of the best scheme on both benchmarks; when out-of-sample errors

Table 6: M4 long series: OWA of each combination scheme (all frequencies), with the number of series with lower / higher MASE than the whole series. *: per-series MASE significantly different from the whole series in either direction (Wilcoxon signed-rank, Bonferroni over 14 schemes). Bold: best per forecaster.

Combination	`AutoARIMA`	`AutoETS`	`ARIMA`(1,0,1)	`ETS`(A,N,N)
Whole series	0.920	0.945	1.056	1.337
Mean	0.975* (2773/2273)	1.027 (2684/2362)	1.196* (2248/2822)	1.336 (1860/1981)
Median	0.905* (2627/1969)	0.926* (2461/2097)	1.064* (2647/2314)	1.336 (1169/1121)
Trimmed mean (20%)	0.956* (2705/2240)	0.927* (2636/2323)	1.179 (2515/2555)	1.336 (1684/1650)
Inverse in-sample MSE	0.912* (2913/2139)	0.943* (2793/2255)	1.056 (2499/2571)	1.336 (1983/2086)
Inverse in-sample relative MAE	0.924* (2874/2172)	0.955* (2715/2329)	1.086 (2359/2711)	1.336 (1859/1987)
Inverse squared in-sample relative MAE	0.906* (2918/2128)	0.939* (2795/2254)	1.059 (2417/2653)	1.336 (1860/1989)
Akaike-style weights ($n_0 = 100$)	0.923* (2825/2199)	0.978 (2597/2412)	1.043* (2658/2407)	1.338 (1931/2040)
Best in-sample window	0.920* (2579/2191)	0.960 (2248/2152)	1.045 (2409/2369)	1.338* (1281/1493)
Inverse out-of-sample MSE	0.910* (2881/2170)	0.926* (2796/2252)	1.047 (2454/2616)	1.336 (2035/2160)
Median of better half	0.898* (2581/1978)	0.924* (2373/2123)	1.030* (2521/2304)	**1.336** (1116/1066)
Mean, $w \geq 300$	0.897* (2929/2055)	0.935* (2773/2210)	1.035* (2671/2398)	1.336 (1695/1618)
Median, $w \geq 300$	0.899* (2605/1937)	0.930* (2422/2071)	1.035* (2685/2212)	1.337 (1039/949)
Inverse in-sample MSE, $w \geq 300$	**0.892*** (2924/2069)	0.938* (2743/2257)	**1.025*** (2749/2319)	1.336 (1851/1799)
Inverse out-of-sample MSE, $w \geq 300$	0.894* (2909/2087)	**0.922*** (2779/2210)	**1.025*** (2653/2415)	1.336 (1776/1744)

Algorithm 1 WINDOWER: fit and forecast with a combination of training windows

Require: forecaster F, series $y_{1:t}$, horizon h, seasonal period m, grid ratio ρ (e.g. 1.5), minimum window $w_{\min} = \max(50, 3m)$, combination $C \in \{\text{median}, \text{median of better half}\}$

1: decide seasonal adjustment on the whole series $y_{1:t}$ (if F is non-seasonal)
2: $W \leftarrow \{w_{\min}\rho^k : w_{\min}\rho^k < t\} \cup \{t\}$ {geometric grid plus the whole series}
3: **for** $w \in W$ **do**
4: fit a clone F_w to $y_{t-w+1:t}$, with the seasonal adjustment of line 1
5: $\hat{y}^{(w)}_{t+1:t+h} \leftarrow$ forecast of F_w
6: **end for**
7: **if** C is median **then**
8: **return** $\text{median}_{w \in W}\, \hat{y}^{(w)}_{t+1:t+h}$ (per step)
9: **else**
10: $W' \leftarrow$ the half of W with the lowest OOS_w (holdout or past one-step errors)
11: **return** $\text{median}_{w \in W'}\, \hat{y}^{(w)}_{t+1:t+h}$
12: **end if**

per window are available (from a holdout, or from previous origins in a rolling deployment), the median of the better half of the windows is as robust and slightly more accurate. Two details follow from the experiments: the seasonal adjustment decision is taken once, on the whole series, so that short windows do not switch it off; and the smallest window is bounded below by a multiple of the seasonal period.

[TODO: Implement as `Windower` in horizon, evaluate as an ordinary forecaster on both benchmarks, and check it reproduces the offline results. Cheaper variants: fewer windows; fixed model structure across windows; incremental updates.]

7 Discussion and limitations

- *When windows help.* The evidence is from real long series with structure that changes over time. The variance cost of windowing on a stable process is not measured here; simulated controls (stationary processes, processes with breaks or drifting parameters) are needed to show when windowing helps and when it costs.[TODO: simulations]

- *Seasonal adjustment.* Windows below 300 change the seasonality decision for hourly series

under the M4 rule, so short-window results on hourly data mix two effects.

- *Minutely wind power.* The largest gains (5 series of one dataset) need inspection before being relied on.[TODO: inspect]
- *Selection noise.* Choices made from 30 one-step errors, or from one M4 holdout, are noisy; the oracle is optimistic, particularly on M4 with one test period per series.
- *Multiple comparisons.* Many windows and schemes were compared on the same test origins; following Rossi and Inoue (2012), the headline results are the schemes defined in advance (median, mean) rather than the best of many.
- *Other forecasters.* Only aeon's implementations were evaluated.

8 Conclusion

For long series, automatic ARIMA and ETS should not be fitted to all the data by default. Fitting the forecaster to a geometric grid of recent windows and taking the median of the forecasts reduced the error of `AutoARIMA` and `AutoETS` by 11–13% on LSSH-99 and their OWA by 2% on the long M4 series, at about four times the cost of one fit. The gains are largest on high-frequency seasonal series and absent on series close to a random walk. Weighting windows by in-sample fit is not reliable; using out-of-sample error to keep the better half of the windows helps slightly. Choosing a single window is noisier than combining. [TODO: revise after the wrapper evaluation and the simulated controls.]

References

Katrin Assenmacher-Wesche and M. Hashem Pesaran. Forecasting the swiss economy using VECX* models: An exercise in forecast combination across models and observation windows. *National Institute Economic Review*, 203:91–108, 2008. ISSN 1741-3036. doi: 10.1177/0027950108089681.

Alexander Banwell and Anthony Bagnall. *Time Series Extrinsic Regression Algorithms for Forecasting Long Time Series with a Short Horizon*, pages 149–164. Springer Nature Switzerland, 2026. ISBN 9783032155351. doi: 10.1007/978-3-032-15535-1_10.

Christoph Bergmeir, Rob J. Hyndman, and José M. Benítez. Bagging exponential smoothing methods using stl decomposition and box–cox transformation. *International Journal of Forecasting*, 32(2):303–312, 2016. ISSN 0169-2070. doi: 10.1016/j.ijforecast.2015.07.002.

Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining*, pages 443–448. Society for Industrial and Applied Mathematics, 2007. doi: 10.1137/1.9781611972771.42.

Vítor Cerqueira, Luís Torgo, Fábio Pinto, and Carlos Soares. *Arbitrated Ensemble for Time Series Forecasting*, pages 478–494. Springer International Publishing, 2017. ISBN 9783319712468. doi: 10.1007/978-3-319-71246-8_29.

Gerda Claeskens, Jan R. Magnus, Andrey L. Vasnev, and Wendun Wang. The forecast combination puzzle: A simple theoretical explanation. *International Journal of Forecasting*, 32(3): 754–762, 2016. ISSN 0169-2070. doi: 10.1016/j.ijforecast.2015.12.005.

Todd E. Clark and Michael W. McCracken. Improving forecast accuracy by combining recursive and rolling forecasts*. *International Economic Review*, 50(2):363–395, 2009. ISSN 1468-2354. doi: 10.1111/j.1468-2354.2009.00533.x.

Liudas Giraitis, George Kapetanios, and Simon Price. Adaptive forecasting in the presence of recent and ongoing structural change. *Journal of Econometrics*, 177(2):153–170, 2013. ISSN 0304-4076. doi: 10.1016/j.jeconom.2013.04.003.

Rakshitha Godahewa, Christoph Bergmeir, Geoffrey I. Webb, Rob J. Hyndman, and Pablo Montero-Manso. Monash time series forecasting archive. In *Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, Melbourne, Australia, 3rd edition, 2021. `https://otexts.com/fpp3/`, section 13.7, "Very long and very short time series".

Rob J. Hyndman and Yeasmin Khandakar. Automatic time series forecasting: The forecast package for r. *Journal of Statistical Software*, 27(3), 2008. ISSN 1548-7660. doi: 10.18637/jss.v027.i03.

Rob J Hyndman, Anne B Koehler, Ralph D Snyder, and Simone Grose. A state space framework for automatic forecasting using exponential smoothing methods. *International Journal of Forecasting*, 18(3):439–454, 2002. ISSN 0169-2070. doi: 10.1016/s0169-2070(01)00110-8.

Rob J. Hyndman, Anne B. Koehler, J. Keith Ord, and Ralph D. Snyder. *Forecasting with Exponential Smoothing: The State Space Approach*. Springer, Berlin, 2008. doi: 10.1007/978-3-540-71918-2.

Atsushi Inoue, Lu Jin, and Barbara Rossi. Rolling window selection for out-of-sample forecasting with time-varying parameters. *Journal of Econometrics*, 196(1):55–67, 2017. ISSN 0304-4076. doi: 10.1016/j.jeconom.2016.03.006.

Victor Richmond R. Jose and Robert L. Winkler. Simple robust averages of forecasts: Some empirical results. *International Journal of Forecasting*, 24(1):163–169, 2008. ISSN 0169-2070. doi: 10.1016/j.ijforecast.2007.06.001.

Andre Jungmittag. Combination of forecasts across estimation windows: An application to air travel demand. *Journal of Forecasting*, 35(4):373–380, 2016. ISSN 1099-131X. doi: 10.1002/for.2400.

Stephan Kolassa. Combining exponential smoothing forecasts using akaike weights. *International Journal of Forecasting*, 27(2):238–251, 2011. ISSN 0169-2070. doi: 10.1016/j.ijforecast.2010.04.006.

Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. The m4 competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1):54–74, 2020. ISSN 0169-2070. doi: 10.1016/j.ijforecast.2019.04.014.

Matthew Middlehurst, Ali Ismail-Fawaz, Antoine Guillaume, Christopher Holder, David Guijo-Rubio, Guzal Bulatova, Leonidas Tsaprounis, Lukasz Mentel, Martin Walter, Patrick Schäfer, and Anthony Bagnall. aeon: a Python toolkit for learning from time series. *Journal of Machine Learning Research*, 25(289):1–10, 2024.

M. Hashem Pesaran and Andreas Pick. Forecast combination across estimation windows. *Journal of Business & Economic Statistics*, 29(2):307–318, 2011. ISSN 1537-2707. doi: 10.1198/jbes.2010.09018.

M. Hashem Pesaran and Allan Timmermann. Selection of estimation window in the presence of breaks. *Journal of Econometrics*, 137(1):134–161, 2007. ISSN 0304-4076. doi: 10.1016/j.jeconom.2006.03.010.

M. Hashem Pesaran, Andreas Pick, and Mikhail Pranovich. Optimal forecasts in the presence of structural breaks. *Journal of Econometrics*, 177(2):134–152, 2013. ISSN 0304-4076. doi: 10.1016/j.jeconom.2013.04.002.

Fridtjof Petersen, Jonas M. B. Haslbeck, Jorge N. Tendeiro, Anna M. Langener, Martien J. H. Kas, Dimitris Rizopoulos, and Laura F. Bringmann. Comparing training window selection methods for prediction in non-stationary time series. *British Journal of Mathematical and Statistical Psychology*, 79(2):341–361, 2026. ISSN 2044-8317. doi: 10.1111/bmsp.70018.

Fotios Petropoulos, Rob J. Hyndman, and Christoph Bergmeir. Exploring the sources of uncertainty: Why does bagging for time series forecasting work? *European Journal of Operational Research*, 268(2):545–554, 2018. ISSN 0377-2217. doi: 10.1016/j.ejor.2018.01.045.

Barbara Rossi and Atsushi Inoue. Out-of-sample forecast tests robust to the choice of window size. *Journal of Business & Economic Statistics*, 30(3):432–453, 2012. ISSN 1537-2707. doi: 10.1080/07350015.2012.693850.

Jeremy Smith and Kenneth F. Wallis. A simple explanation of the forecast combination puzzle. *Oxford Bulletin of Economics and Statistics*, 71(3):331–355, 2009. ISSN 1468-0084. doi: 10.1111/j.1468-0084.2008.00541.x.

James H. Stock and Mark W. Watson. Combination forecasts of output growth in a seven-country data set. *Journal of Forecasting*, 23(6):405–430, 2004. ISSN 1099-131X. doi: 10.1002/for.928.

Jing Tian and Heather M. Anderson. Forecast combinations under structural break uncertainty. *International Journal of Forecasting*, 30(1):161–175, 2014. ISSN 0169-2070. doi: 10.1016/j.ijforecast.2013.06.003.